

ECE/CS/ME 539: Activity 08

Single Layer Neural Networks

Solution

Problem 1: True/False

- (a) The McCulloch-Pitts neuron model uses the threshold function.
True
- (b) The linear regression model is equivalent in its architecture to a single neuron model without an activation function.
True
- (c) The logistic regression model is equivalent in its architecture to a single neuron model with a hyperbolic tangent activation.
False. The logistic regression uses sigmoid activation function.
- (d) $\text{ReLU}(-5) = 0$
True
- (e) $\text{TanH}(-3) = -\text{TanH}(3)$
True. $\text{TanH}(-x) = -\text{TanH}(x)$
- (f) $\lim_{x \rightarrow \infty} \text{Sigmoid}(x) = -1$
False. The range of the sigmoid function is (0,1), so it can't be -1. (It also can be proved that $x = -\ln(2) + i\pi$ when $\text{Sigmoid}(x) = -1$. So x is complex but not real value.)
- (g) $\lim_{x \rightarrow \infty} \frac{d}{dx} \text{Sigmoid}(x) = 1$
False. $\frac{d}{dx} \text{Sigmoid}(x) = \text{Sigmoid}(x)(1 - \text{Sigmoid}(x))$. When x becomes large enough, $\text{sigmoid}(x)=1$ and thus the derivative becomes $1(1 - 1) = 0$.
- (h) Increasing the temperature T of the Sigmoid activation, makes the transition between the "off" ($f(x)=0$) and "on" ($f(x)=1$) states faster.
False. $\frac{d}{dx} \text{Sigmoid}(x/T) = \frac{1}{T} \text{Sigmoid}(x)(1 - \text{Sigmoid}(x))$ where T indicates the temperature. We can see $\frac{d}{dx} \text{Sigmoid}(x/T)$ is smaller as increasing T , that means the transition between the "off" and "on" states is slower.

Problem 2: Derivatives of Activation Functions

Let $f(x) : \mathbb{R} \rightarrow \mathbb{R}$ be an activation function. Compute the derivatives $df(x)/dx$ for the activation functions below.

Activation Function	$f(x)$	$f'(x) = df(x)/dx$
ReLU	$\max(0, x)$	$\begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$
Sigmoid	$\frac{1}{1 + e^{-x/T}}$	$\frac{1}{T} f(x)(1 - f(x))$
Hyperbolic tangent	$\frac{e^{x/T} - e^{-x/T}}{e^{x/T} + e^{-x/T}}$	$\frac{1}{T} (1 - f(x)^2)$
Inverse tangent	$\frac{2}{\pi} \tan^{-1}\left(\frac{x}{T}\right)$	$\frac{1}{T} \frac{1}{1 + (\frac{x}{T})^2}$

Answer: Note that the derivatives of activation functions may be computed directly from x or by reusing $f(x)$. Reusing $f(x)$ is useful because it allows us to compute derivatives using the outputs of the activation function, which must be computed anyway, thus saving on compute.

ReLU:

$$f'(x) = \frac{d}{dx} \max(0, x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

Sigmoid:

$$\frac{df(x)}{dx} = \frac{(1/T)e^{-x/T}}{(1 + e^{-x/T})^2} = f(x) \cdot \frac{1}{T} \cdot \left(1 - \frac{e^{-x/T}}{1 + e^{-x/T}}\right) = \frac{1}{T} f(x)(1 - f(x))$$

Hyperbolic:

$$\frac{d}{dx} \left(\frac{1 - e^{-2x/T}}{1 + e^{-2x/T}} \right) = \frac{(4/T)e^{-2x/T}}{(1 + e^{-2x/T})^2} = \frac{1}{T} \left[1 - \left(\frac{1 - e^{-2x/T}}{1 + e^{-2x/T}} \right)^2 \right] = \frac{1}{T} (1 - f(x)^2)$$

Inverse tangent:

$$\frac{d}{dx} \left(\frac{2}{\pi} \tan^{-1}\left(\frac{x}{T}\right) \right) = \frac{2}{\pi} \cdot \frac{d(\tan^{-1}(x/T))}{d(x/T)} \cdot \frac{d(x/T)}{dx} = \frac{2}{\pi T} \cdot \frac{1}{1 + (x/T)^2}$$

Problem 3: Gradient Descent for Logistic Regression

In this problem, we will walk through a single step of the gradient descent algorithm for logistic regression. As a reminder,

$$\hat{y} = f(x, \theta)$$

$$f(x; \theta) = \sigma(\theta^\top x)$$

$$\text{Cross entropy loss } L(\hat{y}, y) = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

$$\text{The single update step } \theta^{t+1} = \theta^t - \eta \nabla_\theta L(f(x; \theta), y)$$

- (a) Compute the first gradient $\nabla_\theta L(f(x; \theta), y)$.

Solution

We have:

$$\hat{y} = \sigma(\theta^\top x) = \frac{1}{1 + \exp(-\theta^\top x)}$$

Computing the derivative of $\hat{y}$ w.r.t θ and since $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ we get:

$$\frac{d\hat{y}}{d\theta} = \hat{y}(1 - \hat{y}) \frac{d(\theta^\top x)}{d\theta} = (\sigma(\theta^\top x)[1 - \sigma(\theta^\top x)])x$$

Calculating the derivative of the loss function $L(\hat{y}, y)$ w.r.t $\hat{y}$, we get:

$$\frac{dL(\hat{y}, y)}{d\hat{y}} = -\frac{y}{\hat{y}} + \frac{1 - y}{1 - \hat{y}}$$

Applying the chain rule, we find the derivative of $L(\hat{y}, y)$ w.r.t θ :

$$\begin{aligned} \frac{dL(\hat{y}, y)}{d\theta} &= \frac{dL(\hat{y}, y)}{d\hat{y}} \cdot \frac{d\hat{y}}{d\theta} \\ &= \left(-\frac{y}{\hat{y}} + \frac{1 - y}{1 - \hat{y}} \right) \cdot \hat{y}(1 - \hat{y})x \\ &= (-y(1 - \hat{y}) + (1 - y)\hat{y})x \\ &= (\hat{y} - y)x \\ &= (\sigma(\theta^\top x) - y)x \end{aligned}$$

- (b) Now assume a two dimensional input. After including a bias parameter for the first dimension, we will have $\theta \in \mathbb{R}^3$.

Initial parameters: $\theta^0 = [0, 0, 0]$

Learning rate $\eta = 0.1$

data example: $x = [1, 3, 2], y = 1$

Compute the updated parameter vector θ^1 from the single update step.

Solution

We have: $\sigma(\theta^\top x) = (\sigma([0, 0, 0]^\top [1, 3, 2])) = \sigma(0) = \frac{1}{1+\exp(0)} = \frac{1}{2} = 0.5$

Therefore,

$$\begin{aligned}\theta^{0+1} &= \theta^0 - \eta (\sigma(\theta^\top x) - y) x = \\ \theta^1 &= \theta^0 - 0.1(0.5 - 1)[1, 3, 2] \\ \theta^1 &= \theta^0 - 0.1(-0.5)[1, 3, 2] \\ \theta^1 &= [0, 0, 0] + [0.05, 0.15, 0.10] \\ \theta^1 &= [0.05, 0.15, 0.10]\end{aligned}$$

Problem 4: Gradient Descent (Application)

Answer: see Ex_SNM_DAT_torch_sol.ipynb